

SWE 4304: SOFTWARE PROJECT LAB 1

TEAM 4

FINAL
PRESENTATION

✦ FEATHER DB





MEMBERS

WASI OMAR SYED

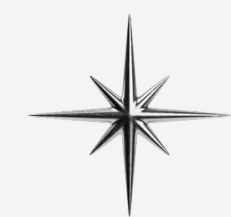
230042143

MD SAIF-AL SARKER

230042106

M SAFWAN HASAN KHAN

230042117



BRIEF OVERVIEW

CUSTOM SQL DATABASE ENGINE

Lightweight SQL implementation
built from scratch

FILE BASED ACCESS CONTROL

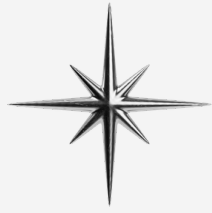
Data is stored on disk using file-
based storage for persistence

3 FUNCTIONAL COMPONENTS

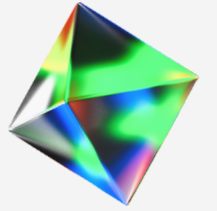
Storage Manager, Query
Processor, Transaction
Manager

TEST DRIVEN DEVELOPMENT

Tests first, code second. In other
words, precision built in from the
start



FEATURES



SQL PARSING & AST

Tokenizes SQL and builds
Abstract Syntax Tree

CRUD OPERATIONS

CREATE, READ,
UPDATE, DELETE
operations

RBAC

Role Based Access
Control

ERD TABLE

Shows relationships between
tables

NESTED QUERIES

Supports subqueries in
SELECT, FROM, WHERE

FILTERING & SORTING

WHERE clause filtering
and ORDER BY sorting

INTERACTIVE REPL

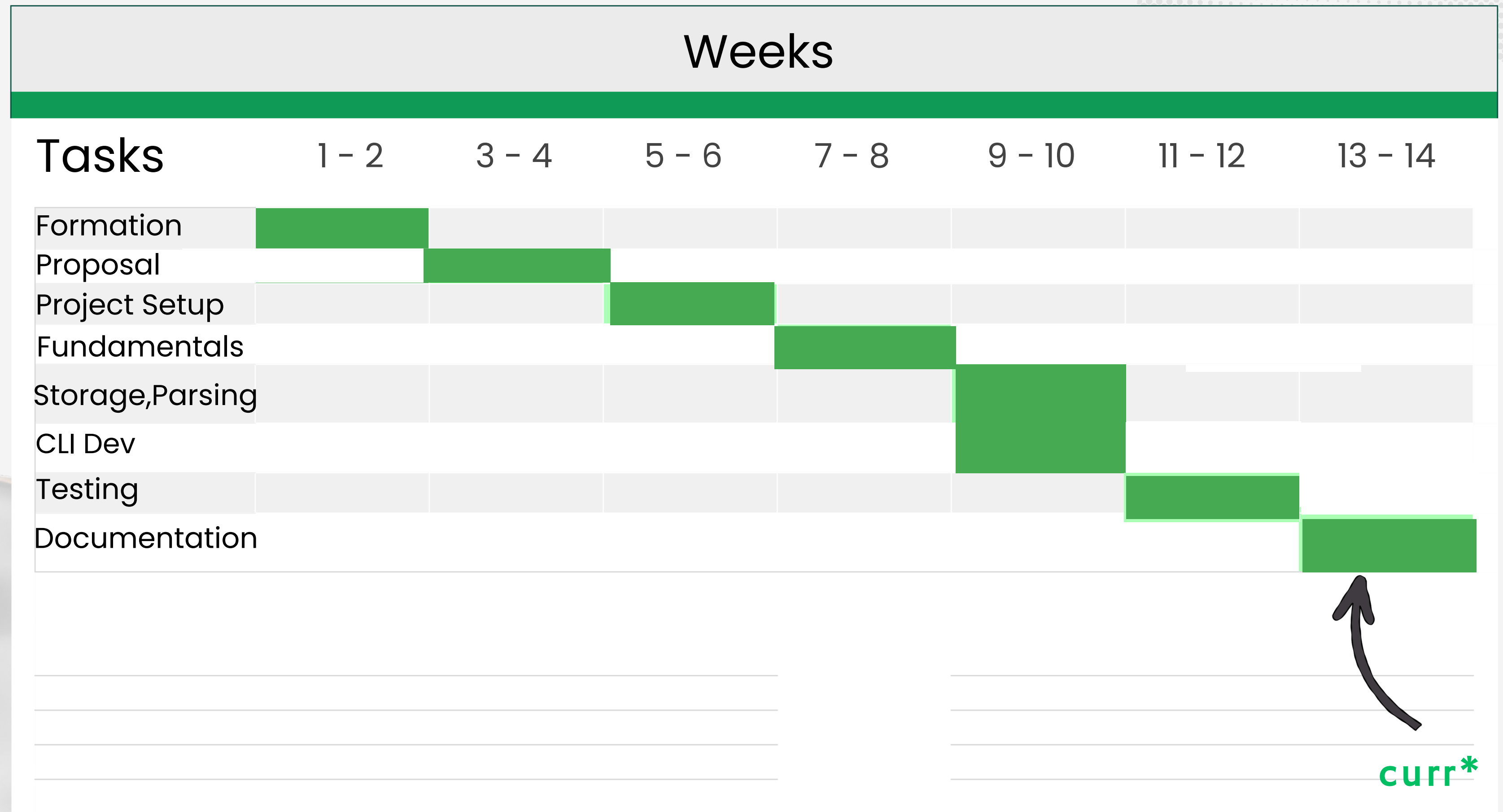
Real-time SQL command
execution and feedback

PERSISTENT STORAGE

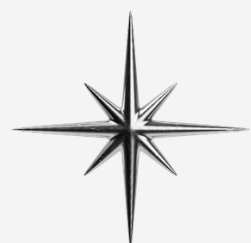
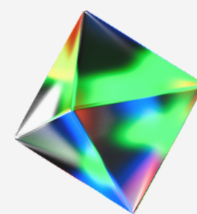
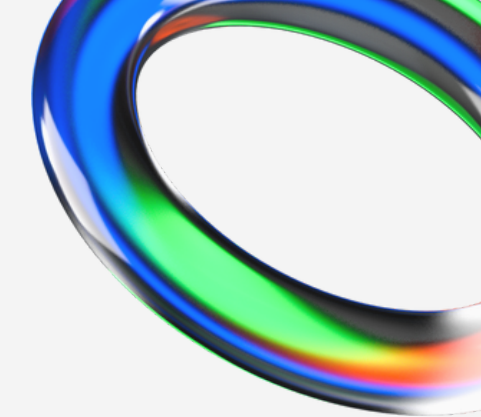
Data survives application
shutdown



TIME LINE



DEMONSTRATION



THANK YOU!

